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The Editors-in-Chief
Knowledge-Based Systems

Re: submission of "The Model Knows the Textbook but Not Your Class: Curriculum Knowledge Graph Constraints for Slide Generation in Two Vocational-Education Courses"

Dear Editors,

On behalf of all authors, I am submitting the manuscript above for consideration as a research article in *Knowledge-Based Systems*. It is a knowledge-representation paper before it is a generation paper: the object is not a better slide generator but *how curriculum knowledge must be represented in order to be actionable by a machine*, and how a machine can then check that the requirement it stated was met. Mapped onto the journal's four expectations:

(1) A knowledge-driven contribution. Course-level knowledge—the concepts a session must cover, the misconception this cohort keeps committing, what must not be introduced yet—is held in a curriculum knowledge graph and compiled, before generation, into an executable *session contract*. The contract is the knowledge representation under test: presence, absence and a corrective slot become hard constraints; order is expressible in the graph but cannot be withheld by the interface, and we report that as the boundary of the representation rather than as a result about ordering.

(2) Methodological novelty. Constraint-as-data: `UnitConstraints` is read twice, once to build the prompt and once to score the finished deck, so what is demanded and what is checked are one object and cannot drift. This differs from retrieval—a retrieved passage decides what the model *reads*, a compiled contract decides what the output *owes*—and it is testable: the design is what let us find seven measurement defects in our own pipeline, each silently favouring the constrained condition, all documented with their fixes.

(3) Isolation of knowledge components. The central comparison is not "text present vs absent" but a 2×2 over the *same* passage. Holding the misconception statement fixed, we cross whether the contract carries a slot for it: 1.000 with structure and text, 0.111 with structure alone, 0.556 with text alone, 0.222 with neither. The interaction is +0.556—the same passage is worth +0.889 inside the contract and +0.334 outside it. One line: the contract does not deliver the text, it makes the text land. A three-arm ablation then prices each block separately: the forbidden-term list buys 0.083 of out-of-graph control and moves nothing else, and the slot prices at 0.889 of misconception coverage.

(4) A real deployment context. Two higher-vocational courses in two languages (Python data structures; C++ arrays), six conditions (54 runs), three ablations (27 runs), a length-matched placebo, a blind review by three teachers of the course (ICC(2,3) = 0.92, ordering agreement with the automatic metrics), the claim-bearing conditions replicated on two further models, and a bounded repair loop that takes the main grid from 6/9 to 9/9 in three calls while the weakest backbone moves only from 4/6 to 5/6.

Sample boundaries, stated up front. The scope claim is analytic: which syllabus dimensions compile into a criterion is a property of the representation and does not rest on the sample size. The effect claim is bounded and reported as such—descriptive intervals, a 2×2 assembled post hoc from conditions run for other purposes and split across two batches, and a five-cell sweep with four runs per cell that separates course from backbone *in sign, not in size*. With three sessions an exact sign-flip test cannot return a two-sided p below 0.25; we say so in the paper and do not dress the result as a significance claim. Nothing here measures learning outcomes.

Reproducibility. Code, both curriculum graphs, prompts, the protocol and all raw outputs are archived in a public repository (Zenodo DOI: `https://doi.org/10.5281/zenodo.XXXXXXX`); every table here and in the supplementary material is regenerated from it by the included scripts. No proprietary or licensed textbook content is redistributed.

Declarations. The manuscript is original, is not under consideration elsewhere, and all four authors have approved the version submitted and agree to its submission here. No competing interests; no specific grant from funding agencies in the public, commercial or not-for-profit sectors. The course data are aggregate classroom statistics; no student-level record is included.

I would be glad to provide any further information the editors require.

Yours sincerely,
Suxing Liu
Corresponding author, on behalf of all authors